Natural Language Processing: Assignment 3

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1 Dataset and Preprocessing

To classify news articles into one of four categories (world, sports, business, and sci/tech), we use the AG News dataset from Hugging Face [Zhang et al., 2015]. This dataset consists of news articles with associated labels, titles, and descriptions. Each class contains 30,000 training samples and 1,900 testing samples, resulting in a total of 120,000 training samples and 7,600 testing samples.

We used the official Hugging Face split (120,000 train and 7,600 test). The training set was split into a new training set (of size 108,000) and a development set (of size 12,000) using stratified sampling with a fixed random seed of 67 to ensure reproducibility. The final split sizes are: train 108,000, dev 12,000, and test 7,600.

Preprocessing. We combined the title and description of each news article into a single text input. We cleaned the text by HTML unescaping to convert HTML entities back to their original characters (like `#39;` to `'`) and removed any remaining HTML tags. We also unescaped LaTeXentities (\\ etc.), collapsed backslashes and normalized whitespace. We converted the labels from 1-based to 0-based indexing (i.e., world: 0, sports: 1, business: 2, sci/tech: 3) for easier handling in the classification models. We combine the title and description into a single text input in the format: `Title: <title> Description: <description>` to provide a clear structure for the models to learn from.

Tokenization. We used BERT Tokenizer, which is based on WordPiece tokenization, to tokenize the text inputs. Similar to BPE, WordPiece is a subword tokenization method that can effectively handle out-of-vocabulary words (OOV) and reduce the vocabulary size while still capturing meaningful subword units. We used the pre-trained BERT tokenizer ($bert-base-uncased$) from Hugging Face, which has a vocabulary size of 30,522 tokens. Both BERT and LSTM models were trained using the same tokenizer to ensure consistency in input representation.

2 Models and Hyperparameters

In this assignment, we compare finetuned BERT with our best LSTM model from Assignment 2 on the AG News classification task.

LSTM model. The LSTM model is based on the Long Short-Term Memory architecture introduced by [Hochreiter and Schmidhuber, 1997]. It reads the input sequence token by token, which makes it suitable for modelling word order and longer dependencies in the text. We use a stacked 2-layer LSTM. To avoid learning from padding tokens, we handle variable-length sequences by skipping over padded positions during

the recurrent computation, so the model only processes the real tokens in each example. For classification, we use the final hidden state of the top LSTM layer as the representation of the whole article. In the bidirectional setting, the model combines information from both directions before making the final prediction. Dropout is applied before the final linear layer. This model architecture was chosen as it was the best performing model in Assignment 2; however, due to the change in tokenization for a fair comparison, we re-tuned the dropout rate and learning rate to ensure a fair comparison with the BERT model. As training the LSTM is not the focus of this assignment, no further details on the training process will be provided, but hyperparameters used for final LSTM training can be seen in Table 1. We found that the LSTM significantly benefited from pooling. We used mean pooling across the sequence length dimension to get a more robust representation of the input text.

BERT model. We use DistilBERT [Sanh et al., 2019], a distilled variant of BERT with 6 layers, hidden size 768, and 12 attention heads (66M parameters). We extract the [CLS] token representation from the final hidden layer as input to a feed-forward classification head. A warmup phase of 500 steps gradually increases the learning rate from zero to stabilize early training. Dropout is applied in the classification head to prevent overfitting.

3 Methods

Hyperparameters. We tuned the BERT model with a grid search over the following hyperparameters: learning rate (1e-5, 2e-5, 3e-5), sequence length (96, 128, 160), and batch size [16, 32], this resulted in 18 configurations. We selected the best configuration based on macro-F1 on the development set.

The LSTM model was manually tuned using five hand-designed experiments each isolating one hyperparameter (early stopping, regularization, sequence length, data scale). We selected the best hyperparameters based on macro-F1 on the development set.

We report accuracy and macro-F1 as our main evaluation metrics, with macro-F1 used as the primary measure for model selection because it gives equal weight to all four classes. In addition, we generated classification reports, confusion matrices, learning curves, and files with misclassified test examples for later error analysis. The test set was only used for final evaluation after model selection.

Shared training settings. Both models were trained with the Adam optimiser. We used gradient clipping with a maximum norm of 1.0 and early stopping with a patience of 2 epochs based on validation loss. The batch size for BERT was set to 16, while for the LSTM it was set to 32 to boost the training times.

Robustness/Slicing Evaluation: Length buckets. We evaluated the models on different length buckets to see how performance varies with input length. We created four buckets based on the distribution of word counts in the training set: shortest (0-34 words), short (35-39 words), long (40-45 words), and longest (46+ words). We then evaluated the models separately on the test examples that fall into each bucket. This analysis allows us to understand if the models perform better on shorter or longer texts, which can reveal insights about their ability to capture context and handle varying input lengths. For instance, if a model performs significantly worse on long texts, it may indicate that it struggles to capture long-range dependencies or that it is overfitting to shorter patterns in the training data.

Robustness/Slicing Evaluation: Keyword Masking Probe. We implemented a keyword masking probe to test how much the models rely on specific keywords for classification. We identified the top 6 most informative keywords per class (24 in total) based on the training data using TF-IDF scores. Then, we

created a list of control words, from the whole training vocabulary, that have the same frequency distribution as the keywords but are not informative for classification (i.e., have the same TF score but low IDF). Then, we created a modified test set where these keywords were masked out (replaced with a special [MASK] token), we did the same for the control words, and we evaluated the models on both masked test sets to see how much their performance drops compared to the original test set. A significant drop in performance on the keyword-masked test set would suggest that the model relies heavily on these keywords for classification, while a smaller drop would indicate that the model is using more contextual information and is less sensitive to specific keywords. The control word masking serves as a baseline to ensure that any observed performance drop is indeed due to the masking of informative keywords rather than just any random words. This analysis helps us understand the robustness of the models to changes in key lexical cues and whether they are learning more generalizable patterns or just memorizing specific terms.

4 Results

4.1 Best hyperparameters

The final hyperparameters for both models are shown in Table 1.

Parameter	**BERT**	**LSTM**
Embedding dim d	—	256
Hidden dim H	768	256
Sequence length	128	128
Learning rate	2e-5	3e-4
Weight decay	0.01	0.01
Dropout	—	0.4
Layers L	6	2
Bidirectional	True	True
Attention heads A	6	—
Classification Head dim	256	—
Classification Head dropout	0.1	—

Table 1: Hyperparameter search space for the BERT and best hyperparameters for the LSTM model.

4.2 Model comparison

The final performance of the best BERT and LSTM models on the development and test sets is shown in Table 2. The fine-tuned BERT model significantly outperforms the LSTM model, achieving an accuracy of 94.86% and macro-F1 of 94.86% on the test set, compared to the LSTM's 90.99% accuracy and 91.00% macro-F1. This demonstrates the strong performance of transformer-based models like BERT on text classification tasks, likely due to their ability to capture complex contextual relationships in the text through self-attention mechanisms. The LSTM model, while still performing well, may struggle to capture long-range dependencies and complex patterns in the text as effectively as BERT, which could explain the performance gap between the two models.

Table 2: Model Comparison (Dev + Test)

Model	**Category**	**Dev**		**Test**	
		Acc	**F1**	**Acc**	**F1**
LSTM	neural	0.9103	0.9106	0.9099	0.9100
DistilBERT (fine-tuned)	transformer	0.9450	0.9450	0.9486	0.9486

4.3 Robustness/Slice Evaluations

4.3.1 Length buckets

As mentioned in Section 3, we evaluated the models on different length buckets; *shortest* (0-34 words), *short* (35-39 words), *long* (40-45 words), and *longest* (46+ words), to see how performance varies with input length. The results are shown in Table 3.

Table 3: Length bucket results on the test set.

Model	**Bucket**	**Count**	**Accuracy**	**Macro-F1**
LSTM	shortest	2144	0.8843	0.8815
LSTM	short	1730	0.9145	0.9145
LSTM	long	1986	0.9285	0.9287
LSTM	longest	1740	0.9155	0.9147
DistilBERT	shortest	2144	0.9324	0.9302
DistilBERT	short	1730	0.9555	0.9555
DistilBERT	long	1986	0.9572	0.9571
DistilBERT	longest	1740	0.9517	0.9501

The results show that both models perform better on longer sequences, with the highest accuracy and macro-F1 scores in the *long* bucket (40-45 words) for both models. However, the fact that the *longest* bucket has decreased performance metrics indicates long-term dependency issues in both models. Though this issue is subtler in DistilBERT, with 0.0055 drop in accuracy and 0.0066 drop in macro-F1 compared to the previous bucket, the LSTM model demonstrates a more significant drop of 0.0130 in accuracy and 0.0140 in macro-F1. This disparity in performance suggests that the LSTM struggles more with sequences longer than 45 words because of its limited capacity to efficiently handle long-range dependencies compared to DistilBERT.

Furthermore, both models had the lowest accuracy and macro-F1 scores in the shortest bucket (0-34 words), likely due to the limited context available in shorter sequences, which makes it harder for the models to accurately classify the news articles.

4.3.2 Keyword Masking Probe

As described in Section 3, we conducted a keyword masking probe to evaluate the reliance of the models on specific class-indicative keywords. The identified top 6 keywords for each class, seen in Table 4, were masked.

Table 4: Top 6 class-indicative keywords per class used for masking.

Class	**Top keywords**
World	afp, iraq, iraqi, killed, minister, president
Sports	cup, game, league, season, team, win
Business	fullquote, oil, percent, prices, reuters, stocks
Sci/Tech	internet, microsoft, music, search, software, space

24 random control words that were frequency-matched to the top keywords over the whole training vocabulary are shown in Table 5.

Table 5: Frequency-matched random control words ($n = 24$) sampled from the global training vocabulary.

Control words
agreed, big, billion, business, computer, earnings, firm, general, help, higher, href, investor, investors, just, london, man, months, network, news, play, quickinfo, red, sports, trade

The results of the keyword masking probe are shown in Table 6, which includes the macro-F1 scores for the regular test set, the keyword-masked test set, and the random control-masked test set. The table also shows the drop in macro-F1 for both the keyword-masked and random control-masked test sets compared to the regular test set.

Table 6: Keyword masking probe results on the test set.

Model	**Regular Macro-F1**	**Keyword-Masked F1**	**Random-Control F1**	**Keyword Drop**	**Random Drop**
LSTM	0.9100	0.8958	0.9090	0.0142	0.0010
DistilBERT	0.9486	0.9396	0.9477	0.0090	0.0009

Both models show a greater drop in macro-F1 when the informative keywords are masked compared to when random control words are masked, demonstrating that both models rely on these keywords for classification. However, the LSTM model experiences a larger drop (0.0142) compared to DistilBERT (0.0090). This suggests that the LSTM model is more dependent on keywords for making predictions, while DistilBERT is less sensitive to the masking of these keywords. Hence, DistilBERT appears to be using more contextual information and learning more generalizable patterns for classification, while the LSTM model may be more prone to memorizing specific terms, which limits its robustness and generalization to unseen data.

5 Robustness and Error Analysis

To better understand errors, we examined the first 20 misclassified test examples saved for both models. Rather than looking at isolated mistakes, we focused on examples that reappeared across multiple models and sequence lengths. This makes it possible to identify systematic weaknesses instead of noise from a single run. Full listings of all misclassified examples are provided in Appendix A.

5.1 Sci/Tech - Business Confusion

The clearest pattern is confusion between *Sci/Tech* and *Business*. For both models, *Business* articles misclassified as *Sci/Tech* is the single most common error type, followed by the exact opposite (*Sci/Tech* articles misclassified as *Business*). This is unsurprising: many technology articles in AG News employ business language such as *market*, *shares*, *workers*, or company names. Especially, LSTM was expected to encounter an increased number of such errors due to its increased reliance on keywords compared to DistilBERT. Examples that recur persistently across both models include "Intel to delay product aimed for high-definition TVs" (App. A.1, Ex. 6; App. A.2, Ex. 10) and "Yahoo! Ups Ante for Small Businesses" (App. A.1, Ex. 7; App. A.2, Ex. 13). The reverse confusion also occurs: Sci/Tech articles mislabeled as Business include "Some People Not Eligible to Get in on Google IPO" (App. A.1, Ex. 2; App. A.2, Ex. 5') and "Promoting a Shared Vision" (App. A.1, Ex. 5, App. A.2, Ex. 7), both of which are misclassified in both models. These systematic errors reveal that the boundary between technological innovation and economic reporting is challenging to distinguish.

5.2 World - Business Confusion

Another recurring pattern is confusion between *World* and *Business*. Articles about politics or international affairs are repeatedly misclassified as *Business* when they include strong economic framing. This appears in both models for "Venezuela Prepares for Chavez Recall Vote" (App. A.1, Ex. 3; App. A.2, Ex. 6) and "Oil prices bubble to record high" (App. A.1, Ex. 9; App. A.2, Ex. 14). In both cases, terms such as *oil*, *market*, and *price* likely dominate topic cues and pull predictions toward *Business*.

This pattern is stronger in LSTM, where additional *World* examples are pushed into *Business*, including "India's Tata expands regional footprint via NatSteel buyout" (App. A.2, Ex. 8), "Google Lowers Its IPO Price Range" (App. A.2, Ex. 17), and "Stock Prices Climb Ahead of Google IPO" (App. A.2, Ex. 19). DistilBERT shows the same tendency but with fewer such cases among the first 20 errors, suggesting that contextual modeling helps somewhat but does not fully resolve this boundary.

We also observe confusion between *World* and *Sports*, especially in Olympics-related coverage. In LSTM, *World* examples such as "Live: Olympics day four" and "U.S. Misses Cut in Olympic 100 Free" are predicted as *Sports* (App. A.2, Ex. 9, Ex. 11), while *Sports* examples are sometimes predicted as *World* (App. A.2, Ex. 12, Ex. 16, Ex. 20). DistilBERT shows the same bidirectional confusion, but less frequently in this sample (App. A.1, Ex. 15, Ex. 18, Ex. 19). These cases indicate that event-driven language can override class-specific cues when news articles mix international and sports framing.

Comparing the two models, the overall error types are highly similar, and many difficult examples are shared. The clearest shared failures remain *Sci/Tech–Business* and *World–Business* boundary cases, which suggests that the main limitation is not architecture-specific but rooted in topical overlap in AG News articles. DistilBERT generally commits fewer extreme keyword-driven mistakes, while LSTM appears more sensitive to dominant lexical cues in short news articles.

Compared with Assignment 2 (CNN and LSTM with BPE tokenization), the error profile is very similar. In both assignments, the dominant confusion is still *Sci/Tech* vs. *Business*, with *World* vs. *Business* as another recurring boundary. This consistency across the modeling techniques suggests that the main source of error is label ambiguity in the dataset. Many news articles combine technological and financial language, or mix political and economic framing. Moving from neural sequence models trained from scratch to a pretrained transformer improves scores, but does not remove these fundamental ambiguities.

Overall, the error analysis suggests that the main challenge is separating articles near the boundary between related topics. The distinction between *Sci/Tech* and *Business* remains the hardest for both models, consistent with the class-wise results in Section 4 where these two categories had the lowest per-class F1 scores.

6 Limitations

This study has a few limitations. First, we only evaluate on AG News, so our conclusions may not transfer to datasets with different writing styles or label structures. Second, our robustness analyses have some limitations: the length bucket analysis is based on word count rather than token count, which may not perfectly capture the model's handling of input length, especially for models with different tokenization strategies. Also, the keyword masking probe relies on TF-IDF to identify important keywords, which may not capture all the nuances of what the model considers important for classification. Additionally, for the LSTM we used mean pooling to fix converging issues with the final hidden state representation, but this may not be the optimal way to represent the sequence for classification, and other pooling strategies could yield different results. Finally, we only compare a single LSTM architecture and a single transformer architecture (DistilBERT), so our conclusions about the superiority of transformers may not hold for all possible architectures or configurations of these models.

References

[Hochreiter and Schmidhuber, 1997] Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8):1735–1780.

[Sanh et al., 2019] Sanh, V., Debut, L., Chaumond, J., and Wolf, T. (2019). Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *ArXiv*, abs/1910.01108.

[Zhang et al., 2015] Zhang, X., Zhao, J., and LeCun, Y. (2015). Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems 28 (NIPS 2015)*.

A Detailed Misclassification Examples

Each example is labeled with its dataset index, gold label, and predicted label for reference during error analysis. This appendix provides a listing of misclassified examples from the test set across all experimental runs.

A.1 DistilBERT

Total misclassified examples: 20

Example 1 *(Index 9: Gold: Sci/Tech, Predicted: World)*

```
Title: Card fraud unit nets 36,000 cards Content: In its first two years, the UK's dedicated card fraud unit, has recovered 36,000 stolen cards and 171 arrests -
↪  and estimates it saved 65m.
```

Example 2 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

```
Title: Some People Not Eligible to Get in on Google IPO Content: Google has billed its IPO as a way for everyday people to get in on the process, denying Wall
↪  Street the usual stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a
↪  new level of public participation. But this isn't the case.
```

Example 3 *(Index 36: Gold: World, Predicted: Business)*

```
Title: Venezuela Prepares for Chavez Recall Vote Content: Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in
↪  world oil market.
```

Example 4 *(Index 47: Gold: Sci/Tech, Predicted: Business)*

```
Title: Dell Exits Low-End China Consumer PC Market Content: HONG KONG (Reuters) - Dell Inc. <DELL.O>, the world's largest PC maker, said on Monday it has left
↪  the low-end consumer PC market in China and cut its overall growth target for the country this year due to stiff competition in the segment.
```

Example 5 *(Index 55: Gold: Sci/Tech, Predicted: Business)*

```
Title: Promoting a Shared Vision Content: As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could
↪  do something about it.
```

Example 6 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

Title: Intel to delay product aimed for high-definition TVs Content: SAN FRANCISCO -- In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.

Example 7 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

Title: Yahoo! Ups Ante for Small Businesses Content: Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more competitive with discounters in the space -- which means that small businesses looking to move online get a sweeter deal through ...

Example 8 *(Index 111: Gold: Sci/Tech, Predicted: Business)*

Title: IBM Buys Two Danish Services Firms Content: IBM said Tuesday it has acquired a pair of Danish IT services firms as part of its effort to broaden its presence in Scandinavia. As a result of the moves, IBM will add about 3,700 IT staffers to its global head count. Financial terms of ...

Example 9 *(Index 120: Gold: World, Predicted: Business)*

Title: Oil prices bubble to record high Content: The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.

Example 10 *(Index 207: Gold: Sci/Tech, Predicted: Business)*

Title: Intuit Posts Wider Loss After Charge (Reuters) Content: Reuters - Intuit Inc. (INTU.O), maker of the No. 1 U.S. tax presentation software TurboTax, on Wednesday posted a wider quarterly loss after taking a goodwill impairment charge during its seasonally weaker fourth quarter.

Example 11 *(Index 240: Gold: Sci/Tech, Predicted: Business)*

Title: Bill Clinton Helps Launch Search Engine Content: Former president Bill Clinton on Monday helped launch a new Internet search company backed by the Chinese government which says its technology uses artificial intelligence to produce better results than Google Inc.

Example 12 *(Index 248: Gold: Sci/Tech, Predicted: Business)*

Title: Caterpillar snaps up another remanufacturer of engines Content: PEORIA - Caterpillar Inc. said Wednesday it will acquire a South Carolina remanufacturer of engines and automatic transmissions, increasing its US employment base by 500 people.

Example 13 *(Index 305: Gold: Sports, Predicted: Business)*

```
Title: Don 't expect Tiger to relinquish his top ranking without a fight Content: They 're calling Ohio a quot;battleground state, quot; one of the two or three
↪ places likely to decide November 's presidential election. On local TV, the Bush and Kerry ads air so frequently that it 's easy to forget it 's Bob Costas
↪ who actually runs the country.
```

Example 14 *(Index 310: Gold: Business, Predicted: Sci/Tech)*

```
Title: Clicking on Profits Content: The latest data from the US Department of Commerce further bolsters what we have all suspected: E-commerce sales are
↪ increasing. Not only might one suspect that consumer confidence has been bolstered since last year, there ...
```

Example 15 *(Index 311: Gold: Sports, Predicted: World)*

```
Title: French Take Gold, Bronze in Single Kayak Content: ATHENS, Greece - Winning on whitewater runs in the family for Frenchman Benoit Peschier, though an
↪ Olympic gold is something new. Peschier paddled his one-man kayak aggressively but penalty free in both his semifinal and final runs on the manmade Olympic
↪ ...
```

Example 16 *(Index 316: Gold: World, Predicted: Business)*

```
Title: Crude Price Spike May Send Gas Higher (AP) Content: AP - Amid soaring crude oil prices, gasoline costs have been dropping. But don't expect that to last,
↪ economists say.
```

Example 17 *(Index 317: Gold: Business, Predicted: Sci/Tech)*

```
Title: Amazon Snaps Up China's Largest Web Retailer (NewsFactor) Content: NewsFactor - Amazon.com (Nasdaq: AMZN) has said it will buy Joyo.com Limited -- a
↪ British Virgin Islands company that operates the largest Internet retail Web site in China -- for US $75 million.
```

Example 18 *(Index 344: Gold: Sports, Predicted: World)*

```
Title: Glory Comes Amid Empty Seats and Closed Shutters Content: HERE in Old Europe, people install shutters outside their windows to keep out the heat, the
↪ pollution, the daylight, the noise. They also lock the shutters tight when they go away on holiday.
```

Example 19 *(Index 355: Gold: World, Predicted: Sports)*

```
Title: Greek weightlifter awaits verdict Content: Greek weightlifter Leonidas Sampanis will find out on Sunday if he is to be stripped of his medal.
```

Example 20 *(Index 363: Gold: World, Predicted: Business)*

```
Title: A Founding Father? Content: Give the guy some credit. Tung Chee-hwa, Hong Kong 's embattled Chief Executive, gets precious little of it from his people
↪ these daysand heaps of
```

A.2 LSTM

Total misclassified examples: 20

Example 1 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

```
Title: Prediction Unit Helps Forecast Wildfires (AP) Content: AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures
↪ and endless charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and
↪ flames will roar.
```

Example 2 *(Index 9: Gold: Sci/Tech, Predicted: Business)*

```
Title: Card fraud unit nets 36,000 cards Content: In its first two years, the UK's dedicated card fraud unit, has recovered 36,000 stolen cards and 171 arrests -
↪ and estimates it saved 65m.
```

Example 3 *(Index 19: Gold: Sci/Tech, Predicted: Business)*

```
Title: Storage, servers bruise HP earnings Content: update Earnings per share rise compared with a year ago, but company misses analysts' expectations by a long
↪ shot.
```

Example 4 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

```
Title: IBM to hire even more new workers Content: By the end of the year, the computing giant plans to have its biggest headcount since 1991.
```

Example 5 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

```
Title: Some People Not Eligible to Get in on Google IPO Content: Google has billed its IPO as a way for everyday people to get in on the process, denying Wall
↪ Street the usual stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a
↪ new level of public participation. But this isn't the case.
```

Example 6 *(Index 36: Gold: World, Predicted: Business)*

```
Title: Venezuela Prepares for Chavez Recall Vote Content: Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in
↪ world oil market.
```

Example 7 *(Index 55: Gold: Sci/Tech, Predicted: Business)*

```
Title: Promoting a Shared Vision Content: As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could
↪ do something about it.
```

Example 8 *(Index 56: Gold: World, Predicted: Business)*

```
Title: India's Tata expands regional footprint via NatSteel buyout (AFP) Content: AFP - India's Tata Iron and Steel Company Ltd. took a strategic step to expand
↪ its Asian footprint with the announcement it will buy the Asia-Pacific steel operations of Singapore's NatSteel Ltd.
```

Example 9 *(Index 79: Gold: World, Predicted: Sports)*

```
Title: Live: Olympics day four Content: Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.
```

Example 10 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

```
Title: Intel to delay product aimed for high-definition TVs Content: SAN FRANCISCO -- In the latest of a series of product delays, Intel Corp. has postponed the
↪ launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing
↪ market for high-definition television displays.
```


Example 11 *(Index 88: Gold: World, Predicted: Sports)*

```
Title: U.S. Misses Cut in Olympic 100 Free Content: ATHENS, Greece - Top American sprinters Jason Lezak and Ian Crocker missed the cut in the Olympic 100-meter
↪ freestyle preliminaries Tuesday, a stunning blow for a country that had always done well in the event. Pieter van den Hoogenband of the Netherlands and
↪ Australian Ian Thorpe advanced to the evening semifinal a day after dueling teenager Michael Phelps in the 200 freestyle, won by Thorpe...
```

Example 12 *(Index 100: Gold: Sports, Predicted: World)*

```
Title: Olympic history for India, UAE Content: An Indian army major shot his way to his country 's first ever individual Olympic silver medal on Tuesday, while
↪ in the same event an member of Dubai 's ruling family became the first ever medallist from the United Arab Emirates.
```

Example 13 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

```
Title: Yahoo! Ups Ante for Small Businesses Content: Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will
↪ make it more competitive with discounters in the space -- which means that small businesses looking to move online get a sweeter deal through ...
```

Example 14 *(Index 120: Gold: World, Predicted: Business)*

```
Title: Oil prices bubble to record high Content: The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New
↪ York, light sweet crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.
```

Example 15 *(Index 141: Gold: World, Predicted: Sci/Tech)*

```
Title: Bush Promotes His Plan for Missile Defense System Content: President Bush, in Pennsylvania, said that opponents of a missile defense system were putting
↪ the nation's security at risk.
```

Example 16 *(Index 148: Gold: Sports, Predicted: World)*

```
Title: St. Louis Cardinals News Content: Right-hander Matt Morris threw seven solid innings, but the Cardinals needed a bases-loaded walk to second baseman Tony
↪ Womack and a grand slam from new right fielder Larry Walker to key a six-run eighth inning for a ...
```

Example 17 *(Index 154: Gold: World, Predicted: Business)*

```
Title: Google Lowers Its IPO Price Range Content: SAN JOSE, Calif. - In a sign that Google Inc.'s initial public offering isn't as popular as expected, the
↪ company lowered its estimated price range to between $85 and $95 per share, down from the earlier prediction of $108 and $135 per share...
```


Example 18 *(Index 155: Gold: World, Predicted: Sci/Tech)*

```
Title: Future Doctors, Crossing Borders Content: Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.
```

Example 19 *(Index 196: Gold: World, Predicted: Business)*

```
Title: Stock Prices Climb Ahead of Google IPO Content: NEW YORK - Investors shrugged off rising crude futures Wednesday to capture well-priced shares, sending
↪ the Nasdaq composite index up 1.6 percent ahead of Google Inc.'s much-anticipated initial public offering of stock. In afternoon trading, the Dow Jones
↪ industrial average gained 67.10, or 0.7 percent, to 10,039.93...
```

Example 20 *(Index 206: Gold: Sports, Predicted: World)*

```
Title: Boro Captain Warns of Duo 's Threat Content: Gareth Southgate has warned Barclays Premiership defences to be wary of Middlesbroughs back-to-form strikers
↪ Mark Viduka and Jimmy Floyd Hasselbaink.
```

B Classification Reports and Confusion Matrices

This appendix provides the full test-set classification report and confusion matrix for each experimental run.

B.1 LSTM

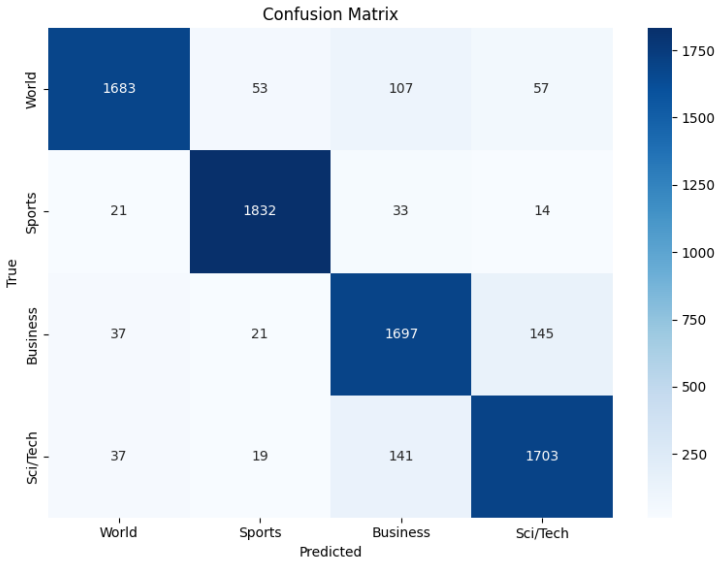

Figure 1: Confusion matrix for LSTM.

```
              precision    recall  f1-score   support

       World     0.9466    0.8858    0.9152      1900
      Sports     0.9517    0.9642    0.9579      1900
    Business     0.8579    0.8932    0.8752      1900
    Sci/Tech     0.8874    0.8963    0.8919      1900

    accuracy                         0.9099      7600
   macro avg     0.9109    0.9099    0.9100      7600
weighted avg     0.9109    0.9099    0.9100      7600
```

B.2 DistilBERT

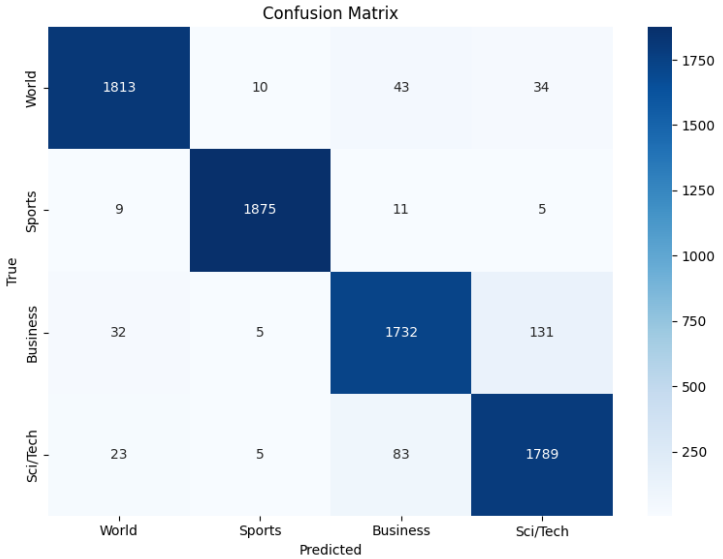

Figure 2: Confusion matrix for DistilBERT with maximum sequence length 128.

```
              precision    recall  f1-score   support

       World     0.9659    0.9542    0.9600      1900
      Sports     0.9894    0.9868    0.9881      1900
    Business     0.9267    0.9116    0.9191      1900
    Sci/Tech     0.9132    0.9416    0.9272      1900

    accuracy                         0.9486      7600
   macro avg     0.9488    0.9486    0.9486      7600
weighted avg     0.9488    0.9486    0.9486      7600
```